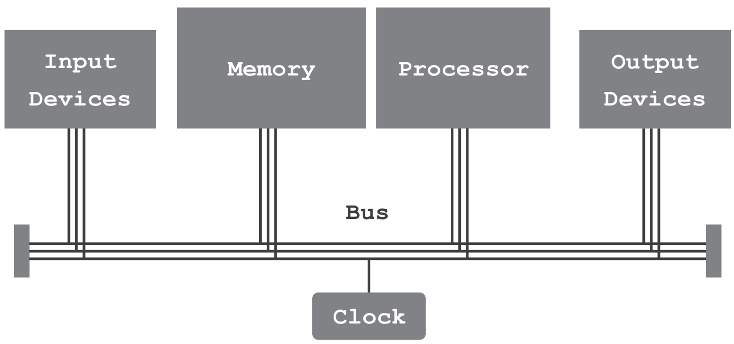
Lesson 5: Design Optimizations

Many considerations are taken when designing a system, all in an effort to design the computer to be fast, dependable, energy efficient, easy to use and affordable. Optimizations target design features, such as performance and dependability. This lesson will focus on the strategies used to optimize performance.

Architecture

All computer systems contain four basic components: input devices, output devices, a processor and a memory. Input devices and output devices allow the exchange of signals with the outside world. The processor handles the conversion of input to output in accordance with stored programs. The memory is tasked with holding information. All information, whether data or instructions, are held in memory.



The basic architecture of a computer includes a processor, memory components, input devices and output devices. In addition, the bus acts as a communication network connecting these components and a clock to ensure all activities are synchronized.

The components of any computer system: a processor, a memory, input devices, output devices, a bus and a clock.

THE BUS SYSTEM

The bus, in computing, is a shared communication pathway that allows the components of a computer system to transmit signals to each other. These signals are divided into 3 groups, each carrying one type of information: data, addresses and control signals. Each of these is transmitted on a dedicated set of conducting lines.

Data bus	The set of lines that transmit the actual data being transferred between components. It is bidirectional, meaning it can transmit data in both directions.
Address bus	The set of lines that transmit the addresses used to specify the source or destination of data.
Control bus	The set of lines that transmit bus control signals that are used to coordinate and control the operations of the components within the computer system.

THE BUS IS A SHARED COMMUNICATION PATHWAY THAT CONNECTS ALL THE COMPONENTS OF A COMPUTER SYSTEM.

In a computer system, each component is connected to the bus and assigned an address, a number that uniquely identifies it. The address allows the components to be identified and accessed by the CPU or other components in the system.

When a component, such as the CPU, wants to communicate with another specific component, such as memory or an I/O device, it places the target component's address on the address bus. This address is then transmitted over the bus to indicate the source or destination of the data transfer.

Example 5.1

Let's say the CPU is busy processing data for streaming a video that is being viewed on the web browser. With the mouse, the user moves the pointer and selects the window for a Word document essay. The screen switches to the essay window, and the video stops playing. What happened inside the computer?

1. User	The user moves the mouse and clicks the left button. This generates signals inside the mouse, which are data signals.
2. Mouse	The mouse prepares the data for transmission and places it on the data bus. It places the CPU address on the address bus. It sends an interrupt signal to the control bus.
3. Bus	The bus, based on the address and control signals, delivers the data to the CPU.
4. CPU	The CPU interrupts the process it is working on, streaming the video, in order to process the incoming data. Based on the new data, the CPU switches the active screen from the web browser to the word processor.

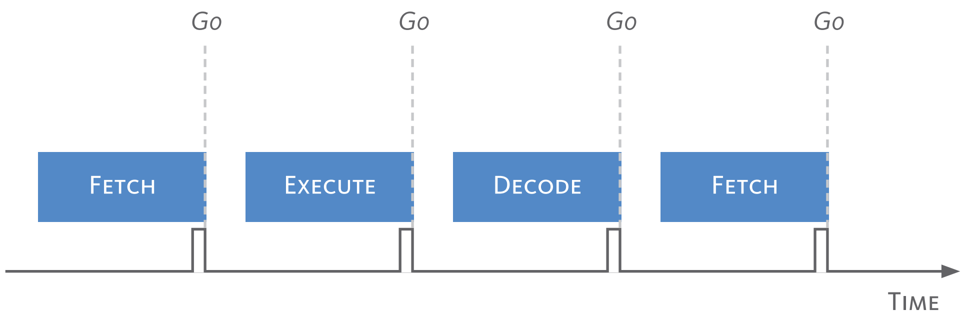
Synchronization

The computer system is a busy place with signals traveling at light speed between the various components while the processor executes one instruction cycle after the other.

Activities occur at different rates. Some components are slower than others, while some instructions take less time to execute than others. This would result in both delays and in errors if left unchecked. Fast operations would pile up, waiting their turn while a slow instruction is being executed, which causes delays and slows performance. Not to mention the errors that can occur if two components try to send data at the same time.

In a synchronized system, all the components operate according to a common time reference, which is typically provided by a time signal generated by an internal clock and distributed to all the components.

The clock has a quartz crystal that oscillates back and forth, vibrates very quickly. Each oscillation is referred to as a clock cycle. With each clock cycle, it generates a timing signal that is placed on the control bus and sent out to all components.



Clock Signal

This timing signal is like a "Go" signal; it tells all the components to move their signals to the next stage. This way, all signals move at the same time and don't "bump" into each other.

Inside the CPU, the clock signals the move from one stage in the instruction cycle to the next. Outside the CPU, the clock signals the transmission of data and addresses on the bus between components. The clock signal is one of the signals transmitted on the control bus.

THE CLOCK GENERATES A TIMING SIGNAL THAT SYNCHRONIZES THE TRANSFER OF ALL SIGNALS THROUGHOUT THE COMPUTER.

The processor operates very quickly, mostly due to the high clock frequency. The clock frequency is the number of clock cycles or oscillations that occur per second. For example, if a clock generates 2 billion clock cycles every second, that means its clock frequency is 2 billion cycles per second, or 2 GHz.

2	Billion	Cycles per second
↓	↓	↓
2	Giga	Hertz
↓	↓	↓
2	G	Hz

Since the clock determines the rate at which the computer operates, then adjusting the clock frequency affects the overall performance of the computer. The clock frequency can be used as a measure of performance, with higher clock frequencies indicating faster performance.

THE CLOCK FREQUENCY IS ONE MEASURE OF COMPUTER PERFORMANCE.

EXAMPLE 5.2

Let's assume there are two hypothetical CPUs, CPU_A and CPU_B. We will make two assumptions for our CPUs, each CPU has 1 processor, and each processor executes only 1 stage of the instruction every clock cycle. If the clock frequency of CPU_A is 6 GHz and that of CPU_B is 3 GHz, then we can say that CPU_A is faster than CPU_B.

$$\frac{6 \times 10^9 \text{ clock cycles}}{1 \text{ second}} \times \frac{1 \text{ instruction}}{3 \text{ clock cycles}} = 2 \times 10^9 \text{ instructions / second}$$

We can also calculate the instruction count per second and compare the two CPUs. If both CPUs are executing 1 stage of the instruction cycle per clock cycle and there are 3 stages, then it will take 3 clock cycles to complete 1 instruction. So CPU $_A$ that runs at 6 GHz, or 6 billion clock cycles per second, will be capable of executing 2 billion instructions per second.

On the other hand, CPU $_B$ is running at 3 GHz or 3 billion clock cycles per second. This is half the speed and is reflected in the calculation, which reveals an instruction count of 1 billion instructions per second.

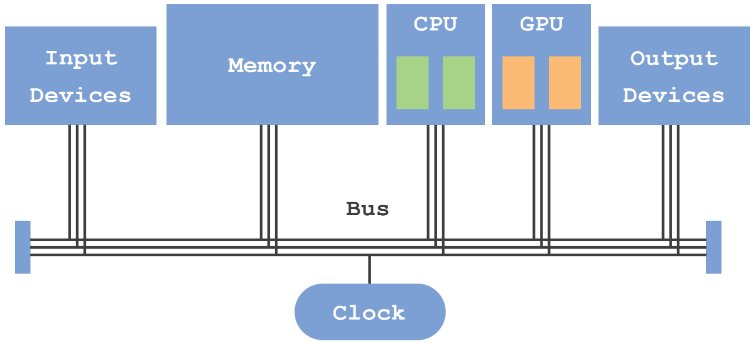
$$\frac{3 \times 10^9 \text{ clock cycles}}{1 \text{ second}} \times \frac{1 \text{ instruction}}{3 \text{ clock cycles}} = 1 \times 10^9 \text{ instructions / second}$$

Parallelism

Another strategy for optimizing performance is referred to as parallelism, which means performing tasks simultaneously at the same time. One way this strategy manifests in the design of the computer is in multi-core CPUs, which are CPUs that have more than 1 processor, each referred to as a core.

Another way this strategy is implemented is in the introduction of the graphic processing unit, GPU. The GPU is a processing unit that focuses only on executing instructions that pertain to graphics. It does not concern itself with running the whole of the computer and its apps. In modern-day computers, particularly those used for games, it is common to have both a CPU and a GPU, and each one of these processing units has multiple cores. This is referred to as parallel architecture.

PARALLELISM REFERS TO THE ABILITY TO PERFORM TASKS SIMULTANEOUSLY.



A parallel architecture that includes a GPU operating in parallel with a CPU. The CPU attends to the operation of the whole computer, while the GPU focuses on processing graphics-related instructions.

It is important to note that while a dual-core CPU has two processors, it is faster than a single-core CPU. However, it does not necessarily mean that it is twice as fast. The overall performance of a computer is not limited to clock frequency; it also depends on the task required.

Example 5.3

Let's assume CPU $_A$ has a single-core, while CPU $_B$ has a dual-core, and both CPUs are operating at the same clock frequency of 2 GHz. Let's also assume that there is a Program A and that the time it takes for CPU $_A$ to execute Program A is 10 seconds.

If CPU B is tasked with executing Program A, its operating system will have to decide whether it can divide Program A into two parts. If it can, then CPU B can execute both parts simultaneously, each on a core. This would shorten the time to execute Program A from 10 seconds to 5 seconds (assuming both parts are equal).

If the operating system cannot divide Program A into two parts, then only one core out of the two will be used, and the execution time remains at 10 seconds.

Proximity

Proximity is another strategy used to optimize computer design. The closer components are together, the better the communication and the faster the transfer of information between them one such implementation of this strategy is the on-chip cache.

Understanding this concept requires an understanding of the memory system and its relation to the processor. Memory is composed of several components that work together as one system, the memory system. The components of this system, in order of size, are: storage, main memory, cache and register file.

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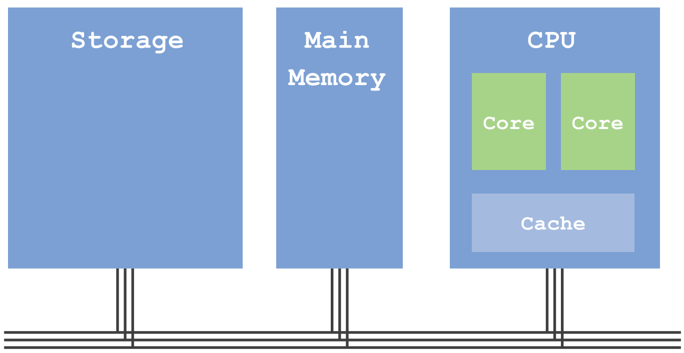


The components of the memory system are the storage, main memory, cache and register file. The register file is not illustrated as it is considered part of the core.

The processor depends on memory to supply it with the information it needs. The execution of tasks requires both instructions and data to be delivered promptly from memory to the processor. The overall performance of the computer depends on memory matching its performance with that of the processor. Memory must keep up with the processor; otherwise, the computer's performance will slow down.

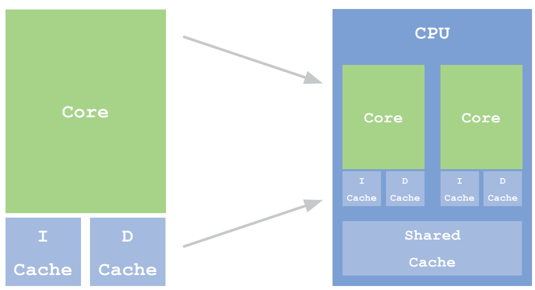
Since close means fast, keeping components, such as the register file and cache, on the same chip as the processor is one way to optimize the performance of the computer. The processor does not have to wait for information to be transmitted on the bus from storage or from main memory. The information it needs is in the registers, and if not there, the information is in the cache. In either case, the information is close by, and the processor does not have to wait.

PROXIMITY OPTIMIZES PERFORMANCE BY PLACING COMPONENTS CLOSE TO THE PROCESSOR.

PREDICTION

Of the optimization strategies, prediction is by far the most ingenious. The ability to predict what comes next greatly improves performance. One such implementation is the separate instruction cache and data cache design.

In this design, the cache is divided into two caches: an instruction cache and a data cache. Having a mixed cache requires the processor to sift through large amounts of information in search of the next instruction. However, we know that instructions follow each other, and therefore, we can predict that if instruction 1 is executed, it will be followed by instruction 2. So, why not arrange all the instructions in order and separate them from the clutter of data?



Each core has two separate caches: one for instructions and the other for data, labeled I-Cache and D-Cache, respectively. The CPU also has a shared cache for all cores on the chip.

Predicting the next instruction allows us to arrange instructions in order and separate them from data. This makes finding the next instruction easy and fast.

PREDICTION OPTIMIZES PERFORMANCE BY PLACING THE MOST LIKELY INFORMATION TO BE USED CLOSEST TO THE PROCESSOR.

Another implementation of prediction is prefetching. When the processor searches for data, it starts by checking the closest memory component to it: the data cache. If it doesn't find the desired data there, it proceeds to check the next component, the shared cache, the main memory, and finally, storage. What's interesting is that when the processor does find the desired data, it doesn't stop at fetching just that particular data item. Instead, it also fetches the adjacent data items. The rationale behind this approach is that if one data item is needed, there is a high chance that the next adjacent data item will be needed soon as well.

By prefetching data that is predicted to be used soon, the processor increases its chances of having the required information available nearby. This proactive data fetching significantly improves the performance of the computer system.

Notes

The principle of locality is fundamental to the operation of the computer. While not explicitly stated as a principle, the lesson clearly identifies both temporal and spatial locality as concepts.

Summary

The components of any computer system are: a processor, a memory, input devices, output devices, a bus and a clock.

The bus is a shared communication pathway that connects all the components of a computer system. The bus is divided into the address bus that transmits source and destination addresses, the control bus that transmits control signals, including the clock timing signal, and the data bus that transmits the actual data.

The clock generates a timing signal that synchronizes the transfer of all signals throughout the computer. Synchronization is a strategy to optimize performance by ensuring that all signals travel without congestion or delay.

The clock frequency is one measure of computer performance. Its unit of measurement is the Hertz. Other measures of performance are the instruction count per second and the program execution time.

Parallel processing refers to the ability to perform tasks simultaneously. Parallelism is a strategy that is implemented in the form of multi-core processors and in the use of graphics processing units that work in parallel with central processing units.

The components of the memory system are: storage, main memory, cache and register file.

Proximity optimizes performance by placing components close to the processor. The closer components are, the faster the exchange of signals. The inclusion of the cache on the same chip as the processor is one implementation of the proximity concept.

Prediction optimizes performance by placing the most likely information to be used closest to the processor. The ability to predict what comes next significantly improves performance. The separation of the instruction and data caches is one implementation of the strategy of prediction. The other is referred to as prefetching, where the processor fetches not only the requested data item but also the adjacent data items. If one data item is needed, there is a high chance that the next adjacent data item will also be needed soon.

Activity 5.1

Read the following statements, and write on the line provided for each statement 'T' if it is true and 'F' if it is false.

T/F	STATEMENT
T	The bus is a component of the computer system.
T	The clock sends out a timing signal to synchronize all components in the computer system.
F	Synchronization is not essential for the proper functioning of the computer..
F	The data bus and the address bus are the only two bus lines.
F	The execution time is not a measure of performance.
T	Performing tasks simultaneously is referred to as parallelism.
T	The CPU and GPU work in parallel.
T	Placing the cache on the same chip as the processor is one implementation of the proximity principle.
T	Prediction is a fundamental concept in computing.
F	Prediction does not improve performance.

Activity 5.2

Complete the sentences below.

1. The ________ is a system of lines that connects all computer components.
   a) bus
   b) processor
   c) cache

2. The clock cycle is the time duration between two ________ .
   a) frequencies
   b) instructions
   c) timing signals

3. The ________ is a measure of computer performance.
   a) instruction count
   b) execution time
   c) mean time to failure

4. The unit of clock frequency is the ________ .
   a) Second
   b) Giga
   c) Hertz

5. The unit of clock cycle is the ________ .
   a) Second
   b) Meter
   c) Hertz

Activity 5.3

Answer the questions below.

1. The clock is an implementation of which concept?
   a) synchronization
   b) parallelism
   c) prediction

2. The cache is an implementation of which concept?
   a) parallelism
   b) proximity
   c) synchronization

3. The separate I-cache and D-cache is an implementation of which concept?
   a) parallelism
   b) prediction
   c) synchronization

4. The multi-core processor design is an implementation of which concept?
   a) parallelism
   b) proximity
   c) prediction

5. The strategy of prefetching is an implementation of which concept?
   a) prediction
   b) synchronization
   c) proximity

Activity 5.4

Answer the questions below.

1. Assuming the CPUs have the same number of cores, which CPU is the fastest?
   a) CPU_A : 3.5 GHz
   b) CPU_B : 3.1 GHz
   c) CPU_C : 2.7 GHz

2. Assuming the CPUs operate at the same clock frequency, which CPU is the fastest?
   a) CPU_A : dual core
   b) CPU_B : octa core
   c) CPU_C : quad core

3. Where is the first place the processor will look for instructions?
   a) D-cache
   b) I-cache
   c) Shared cache

4. Where is the first place the processor will look for data?
   a) Main memory
   b) Shared cache
   c) D-cache

5. If the processor doesn't find data in main memory, where will it look next?
   a) Storage
   b) Shared cache
   c) Register file

Activity 5.5

Answer the questions below.

1. Write the name and specifications of your processor.

   Name: Intel Core i5

   Cores: Quad-Core

   Clock: 3 GHz

2. List, in order, the memory components the processor checks when looking for data. Assume it has already checked the register file.

   D-cache, shared cache, main memory, storage

3. What is prefetching, and why does it work?

   Prefetching is a process by which the processor retrieves the data it is looking for and then fetches it along with adjacent data items. This is a strategy to improve performance. It is effective because if one data item is needed, there is a high chance that the next adjacent data item will be needed soon as well.

Activity 5.6

Measure your computer's execution time. Type the program below, presented in Python. The program tasks the CPU with adding a long list of numbers and measures the time it takes the CPU to complete the task.

Code	Explanation
`import time`	Imports the time module.

`def cpu_intensive_task():`
`        result = 0`
`    for i in range (1, 1000000):`
`        result = result + i`
`    return result`
	Defines a function by the specified name. The function adds all the numbers in range between 1 and 1,000,000.

`start_time = time.time()`	Starts the timer.
`result = cpu_intensive_task()`	Runs the function.
`end_time = time.time()`	Stops the timer.
`execution_time = end_time - start_time`	Calculates the execution time.
`print ("Result:", result)`	Prints the sum.
`print ("Execution Time:", execution_time, "seconds")`	Prints the execution time.

1. Run the program for different ranges of numbers and document the execution time for each.

   Execution time for adding 1 to 1,000,000 : ______________________

   Execution time for adding 1 to 10,000,000 : ______________________

   Execution time for adding 1 to 100,000,000 : ______________________

   ____

   ____